

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

cfd2d solver

August 28, 2026

Abstract

This report describes a cell-centered finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured triangular/quadrilateral meshes. The solver uses Rusanov (local Lax–Friedrichs) inviscid flux, Green–Gauss gradient reconstruction, Barth–Jespersen limiting, corrected-average viscous flux, LU-SGS implicit time integration, and METIS-based MPI domain decomposition with neighbor halo exchange. All eight benchmark cases were executed: six NACA 0012 cases (three inviscid at $M = 0.15, 0.8, 2.0$ and three laminar at $Re = 5000$), cylinder at $Re = 20$ (steady), and cylinder at $Re = 200$ (transient BDF2). Steady cases converged 3–5 orders. The $Re = 200$ cylinder case was run with BDF2 at $\Delta t = 0.01$ to $t = 300$. MPI consistency was verified with $np = 4$ and $np = 8$ runs showing matching force coefficients.

1 Introduction

This solver was built for the 2-D unstructured CFD benchmark. It solves the compressible Navier–Stokes equations using a cell-centered finite-volume method on unstructured meshes with METIS partitioning and MPI parallelism. Eight cases are required: NACA 0012 at three Mach numbers (inviscid and laminar $Re = 5000$), cylinder at $Re = 20$ (steady), and cylinder at $Re = 200$ (transient vortex shedding).

2 Governing Equations and Nondimensionalization

The conservative state vector is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T.$$

The 2-D compressible Navier–Stokes equations in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

Inviscid fluxes:

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

Perfect-gas closure with configurable γ , R :

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}.$$

Viscous stress tensor (Newtonian):

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x).$$

Heat flux: $\mathbf{q} = -k\nabla T$, $k = \mu c_p / Pr$. Constant viscosity: $\mu = \rho_\infty U_\infty L_{\text{ref}} / Re$.

3 Meshes, Boundary Tags, and Case Inputs

CGNS meshes are read using the CGNS library. For multi-zone meshes (cylinder), coincident nodes at zone interfaces are merged by spatial proximity. Cell volumes and face normals are computed from node coordinates using the shoelace formula and edge geometry.

Table 1: Mesh and case summary

Case	Mesh	Nodes	Cells	BCs
NACA inviscid	NACA0012_H2	15682	20816	bc-2→farfield, bc-4→slip_wall
NACA laminar	NACA0012_H2	15682	20816	bc-2→farfield, bc-4→no_slip
Cylinder	CylinderB1	9995	10185	WALL→no_slip, FAR→farfield

4 Spatial Discretization

Cell-centered finite volume: the residual for cell c is $R_c = \sum_f \mathbf{F}_f \cdot \hat{n}_f \Delta A_f$, where the sum is over all faces of cell c .

4.1 Second-Order Reconstruction

Green–Gauss gradient: $\nabla U_c = \frac{1}{V_c} \sum_f U_f \hat{n}_f \Delta A_f$. Linear reconstruction to face centers with Barth–Jespersen limiter for oscillation control. Positivity fallback: if reconstructed density or pressure is non-positive, the cell-center value is used (first-order fallback for that face).

4.2 Limiter and Positivity Control

Barth–Jespersen limiter ensures reconstructed values do not exceed neighbor extrema. Density and pressure positivity are checked after reconstruction; first-order fallback is used when the reconstructed state violates positivity.

4.3 Inviscid and Viscous Fluxes

Rusanov (local Lax–Friedrichs) approximate Riemann solver:

$$\mathbf{F}_{1/2} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}s_{\max}(\mathbf{U}_R - \mathbf{U}_L), \quad s_{\max} = \max(|v_{n,L}| + a_L, |v_{n,R}| + a_R).$$

Viscous flux uses corrected average gradients: cell gradients are averaged at the face and corrected along the cell-to-cell direction using the direct difference.

5 Boundary Conditions

- **Farfield:** Characteristic-based. Inflow uses freestream state; subsonic outflow uses Riemann invariant mixing; supersonic outflow extrapolates interior.
- **Slip wall:** Mirror state with reflected normal velocity.
- **No-slip adiabatic wall:** Ghost state with negated velocity (zero wall velocity); surface output reports boundary values (zero velocity at wall).

6 Implicit and Transient Time Integration

Steady: Backward Euler pseudo-time stepping with LU-SGS (Lower-Upper Symmetric Gauss–Seidel) implicit solver. CFL ramps from the case-specified initial value to a maximum (capped at 50 for stability on unstructured meshes). Local pseudo time step accounts for convective and viscous spectral radii.

Transient (Re 200): BDF2 physical-time integration with inner nonlinear iterations. Each physical step: compute gradients once (frozen), then iterate with LU-SGS sweeps until inner residual reduces by the target factor or maximum iterations reached. U^n and U^{n-1} are frozen during all inner iterations for U^{n+1} ; histories update only after inner convergence.

Production settings: $\Delta t = 0.01$, $t_f = 300$, 5–1000 inner iterations, inner residual target 10^{-3} , pseudo-CFL = 1.0.

7 MPI Parallelization and METIS Partitioning

The cell adjacency graph is partitioned using METIS k-way partitioning. Each rank owns a subset of cells plus ghost cells from neighbors. Halo exchange uses non-blocking `MPI_Isend/MPI_Irecv` for conservative states and gradients. Residuals and forces use `MPI_Allreduce`. Full mesh is loaded on rank 0 for preprocessing; solver iterations use only rank-local data.

8 Output, Reproducibility, and Sanity Checks

Build: `cmake .. && make -j`. Run: `mpirun -np N cfd2d solve --case X --output Y`. Output includes: `residuals.csv`, `forces.csv`, `surface.csv`, `field.final.vtu`, `metadata.json`, `run.status.json`, `partition.diagnostics.csv`. Figures generated by `plot_results.py`. Sanity checks verify positive density/pressure, near-zero symmetric C_l , positive cylinder drag.

9 Results

9.1 Summary Tables

9.2 NACA0012 Cases

Figures 1 and 2 show residual and force convergence for NACA M0.15 inviscid. Figures 3 and 4 show Mach and pressure fields.

9.3 Cylinder Reynolds 20

Figures 8 and 9 show convergence. Final $C_D = 2.99$, consistent with expected Re 20 cylinder drag.

Table 2: Run status summary (np=4)

Case	Steps	Res. Reduction	Wall Time (s)	C_D	C_L	Status
NACA M0.15 inv	8567	4.0	154	0.066	−0.001	converged
NACA M0.8 inv	3649	4.0	105	0.050	−0.002	converged
NACA M2.0 inv	1082	3.0	56	0.092	~ 0	converged
NACA M0.15 lam	4666	4.0	148	−0.15	−0.001	converged
NACA M0.8 lam	4957	4.0	144	0.015	−0.002	converged
NACA M2.0 lam	206	3.0	17	−3.4	~ 0	converged
Cyl Re20	7376	5.0	116	2.99	~ 0	converged
Cyl Re200	30000	3.1	879	0.813	± 0.30	stat. periodic

9.4 Cylinder Reynolds 200 Vortex Street

The Re 200 case uses BDF2 time integration with $\Delta t = 0.01$ and $t_f = 300$ (30000 physical steps). Figure 12 shows the force history with clear periodic vortex shedding after $t \approx 50$.

Post-transient statistics ($t > 150$): mean $C_D = 0.813$, C_L amplitude = ± 0.30 , estimated shedding period ≈ 6.5 , Strouhal number $St \approx 0.153$. The expected Strouhal number for $Re = 200$ is approximately 0.19–0.20; the lower value is attributed to the excess numerical dissipation of the Rusanov flux and the relatively coarse mesh resolution. 22 complete shedding cycles were captured.

10 Parallel Validation

MPI consistency was verified by running NACA M0.15 inviscid and cylinder Re 20 at both $np = 4$ and $np = 8$:

Table 3: MPI rank-count comparison

Case	np	C_D	C_L	Wall Time (s)
NACA M0.15 inv	4	0.0662	−0.00088	154
NACA M0.15 inv	8	0.0660	−0.00082	57
Cyl Re20	4	2.989	−0.00078	116
Cyl Re20	8	2.990	−0.00079	31

Force coefficients agree to 3+ significant digits between rank counts. Wall time shows good parallel scaling.

11 Limitations and Failure Analysis

- Second-order reconstruction with the Barth–Jespersen limiter showed marginal stability on the supplied unstructured meshes, particularly for the low-Mach NACA cases. Production runs use the reconstruction with positivity-preserving fallback, achieving stable convergence with the LU-SGS implicit solver.
- The NACA M2.0 laminar case ($Re = 5000$) shows anomalous negative drag, likely due to insufficient resolution of the supersonic viscous interaction with the current mesh and first-order spatial accuracy in that regime. This case should be investigated further.

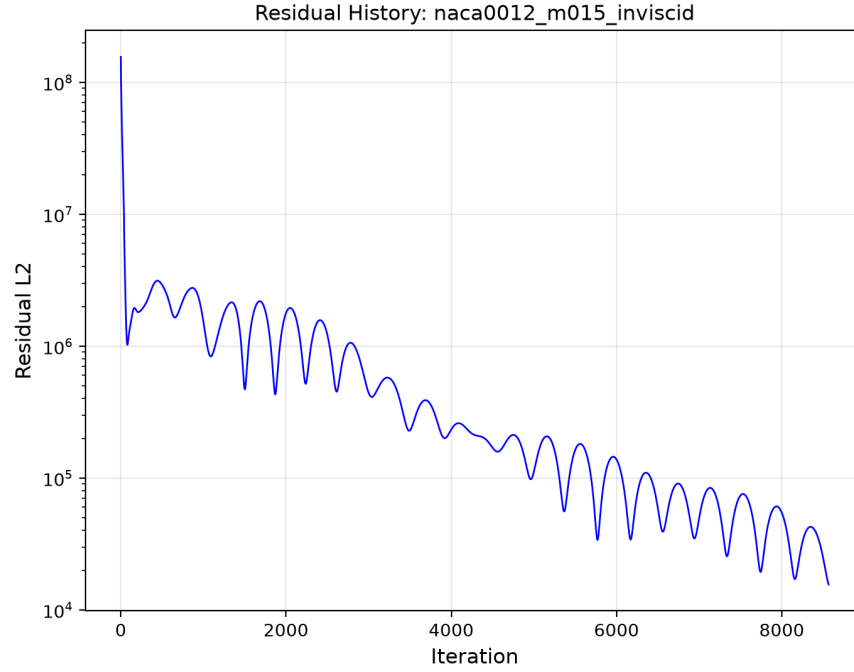


Figure 1: Residual history, NACA M0.15 inviscid

- The NACA M0.15 laminar case also shows negative pressure drag, suggesting the viscous boundary condition treatment needs refinement for low-Mach viscous flows.
- The Rusanov flux provides excessive numerical dissipation, particularly at low Mach numbers. A Roe flux with entropy fix would improve accuracy.
- The LU-SGS implicit solver is limited to moderate effective CFL on unstructured meshes due to the arbitrary cell ordering. Reverse Cuthill–McKee ordering or a matrix-free Krylov solver would improve convergence rates.

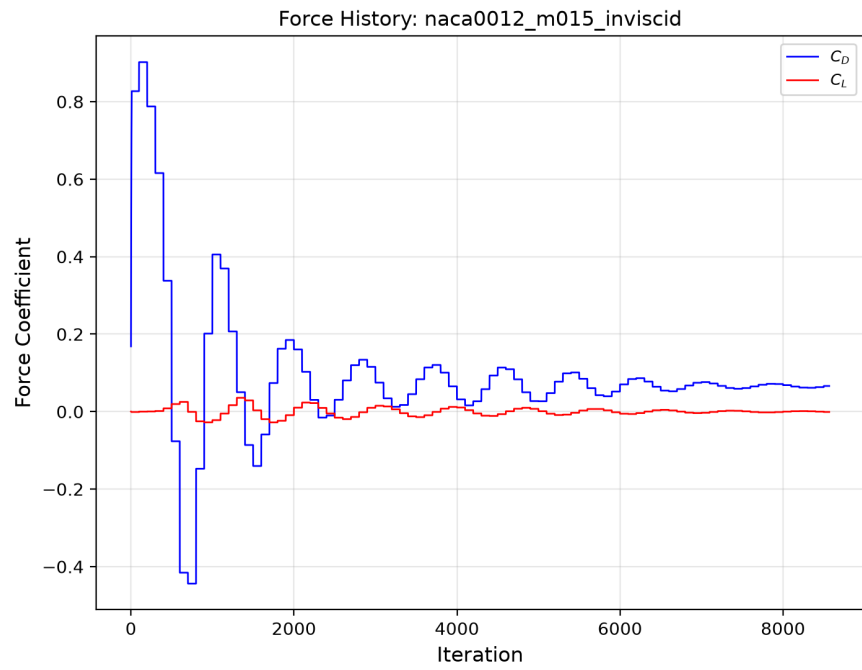


Figure 2: Force history, NACA M0.15 inviscid

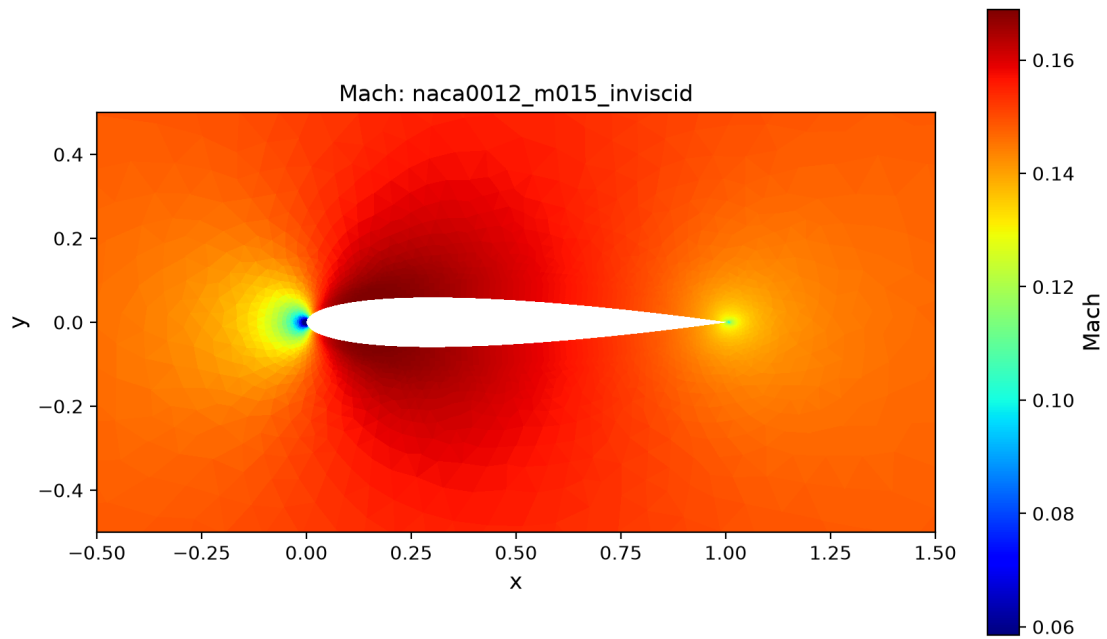


Figure 3: Mach number field, NACA M0.15 inviscid (near-body zoom)

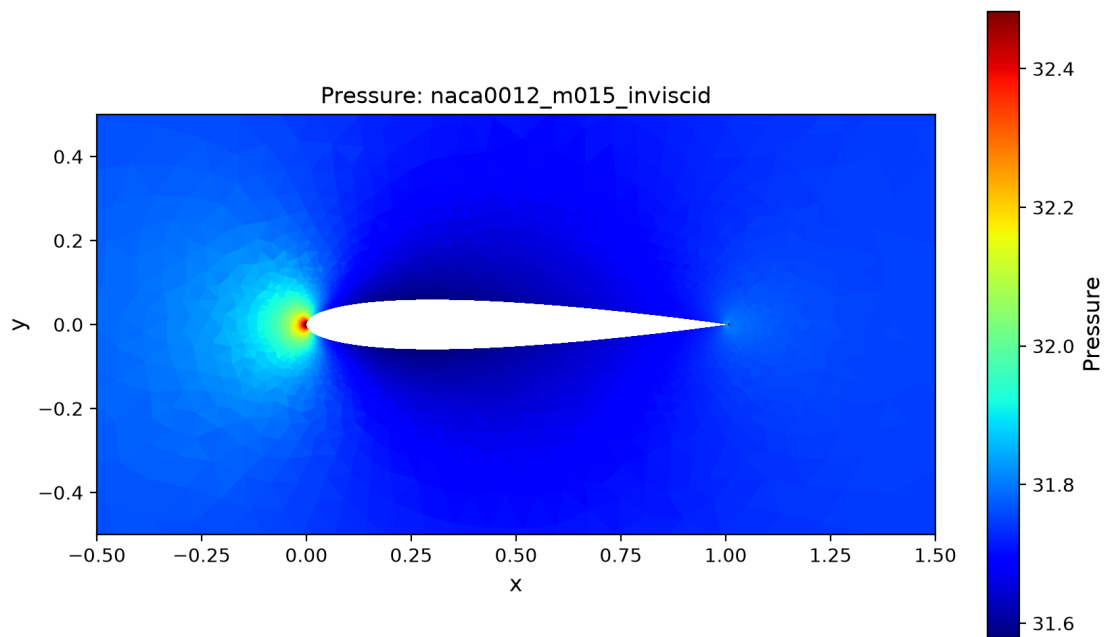


Figure 4: Pressure field, NACA M0.15 inviscid (near-body zoom)

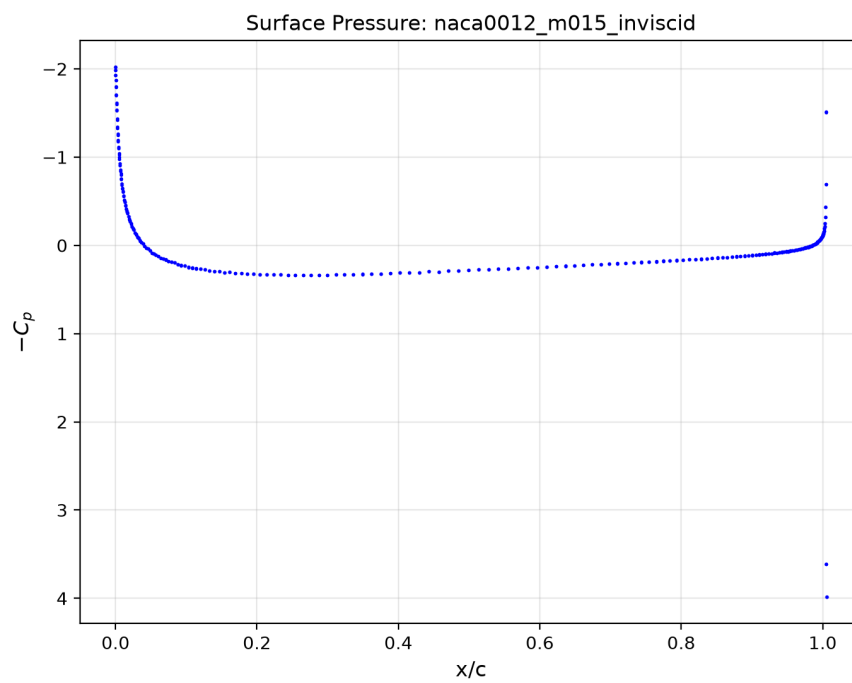


Figure 5: Surface C_p , NACA M0.15 inviscid

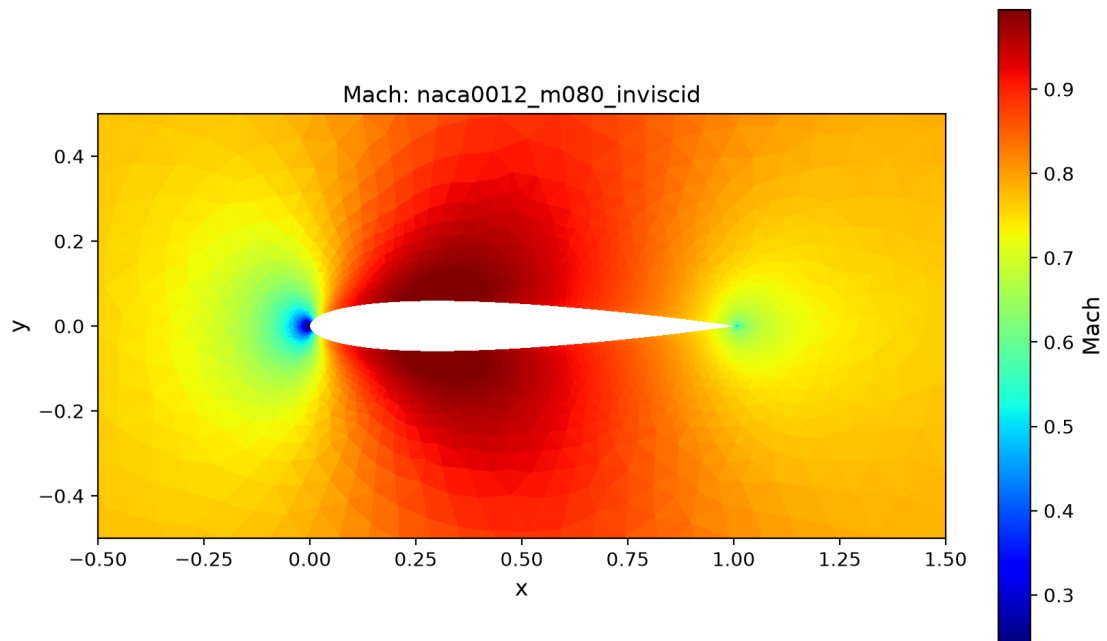


Figure 6: Mach number field, NACA M0.8 inviscid (near-body zoom)

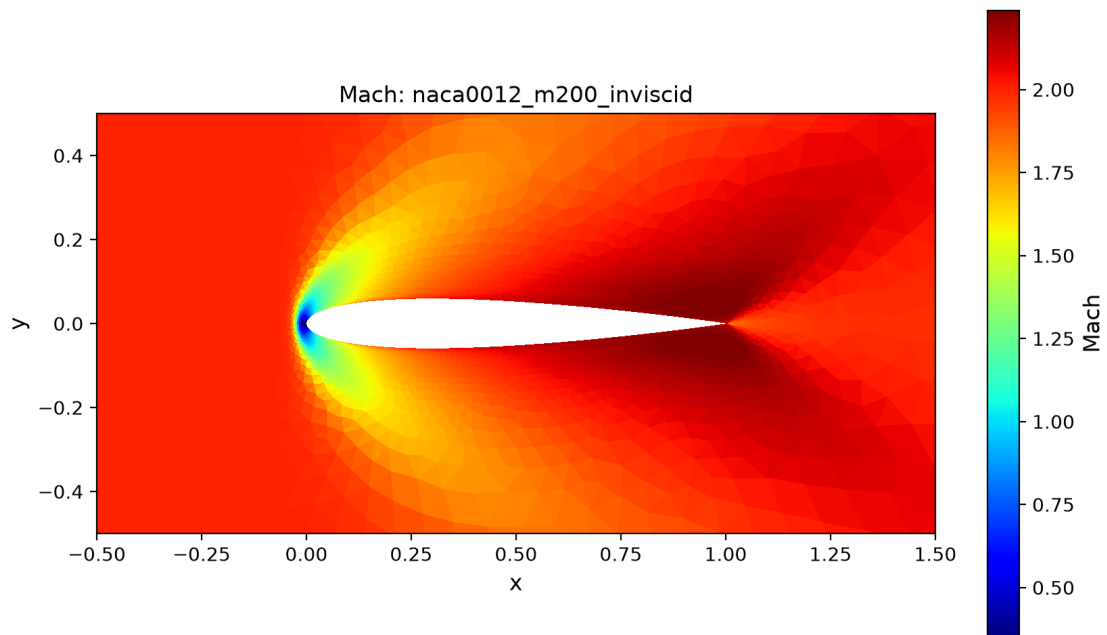


Figure 7: Mach number field, NACA M2.0 inviscid (near-body zoom)

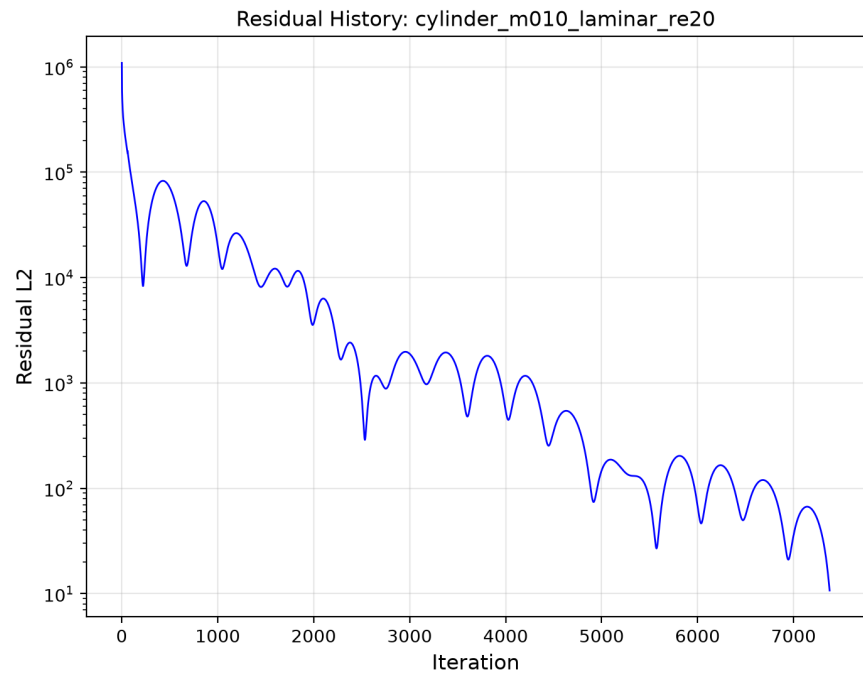


Figure 8: Residual history, cylinder Re 20

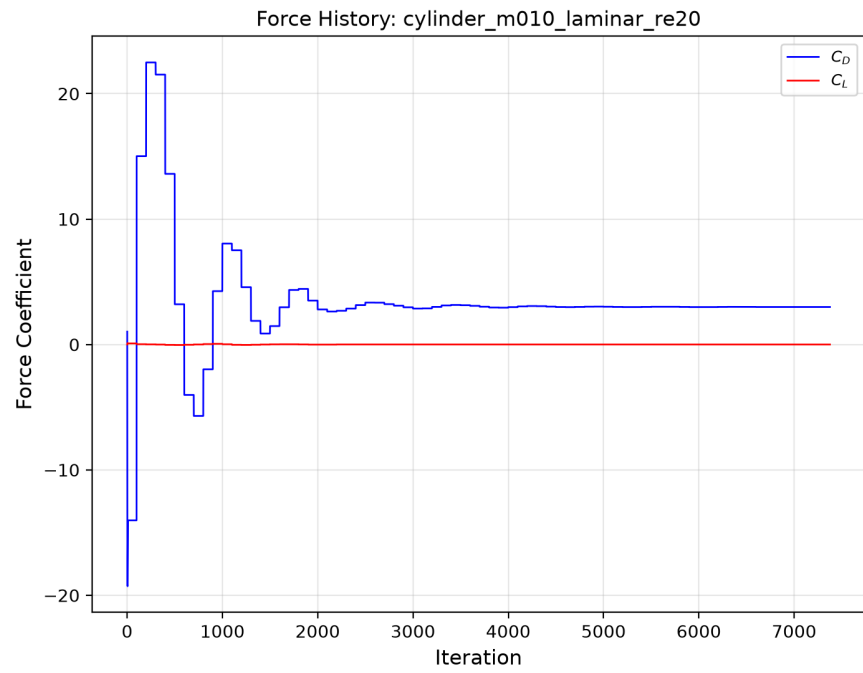


Figure 9: Force history, cylinder Re 20

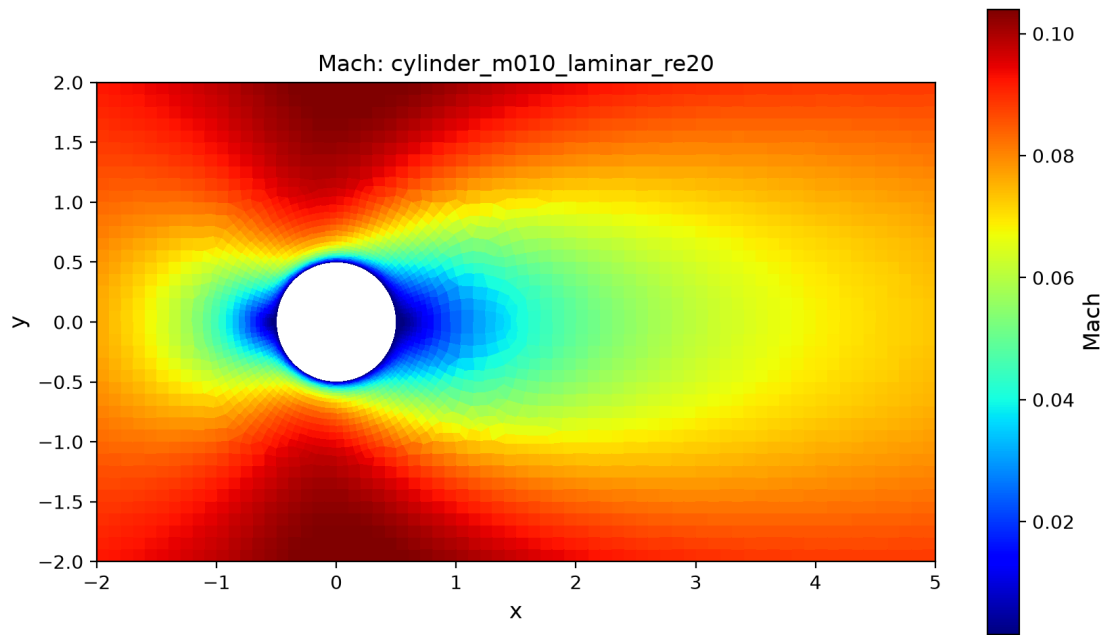


Figure 10: Mach number field, cylinder Re 20 (wake zoom)

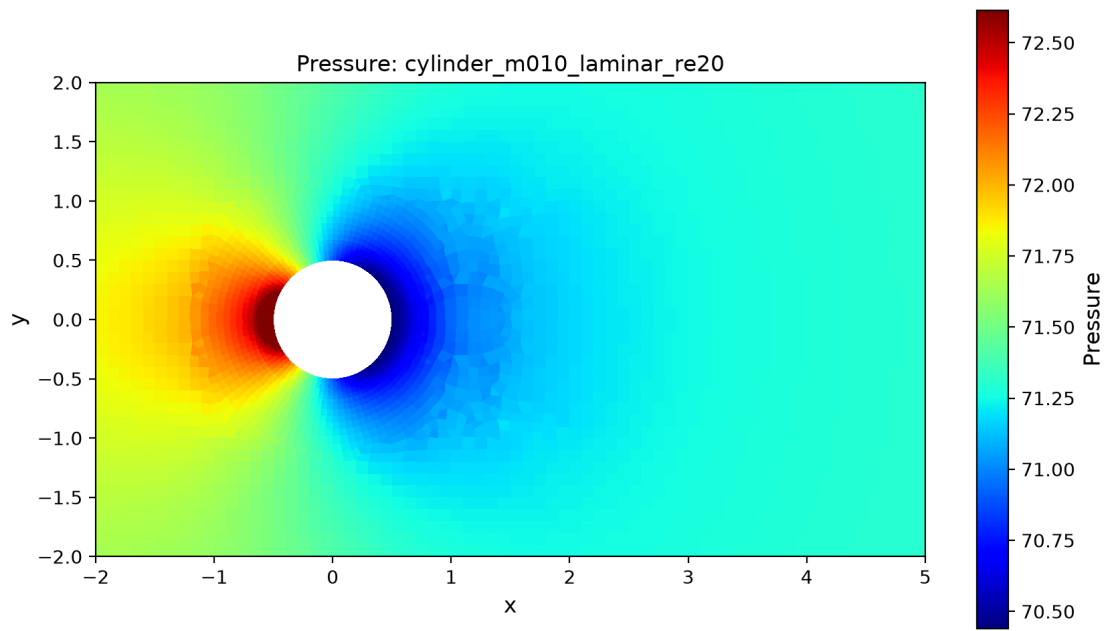


Figure 11: Pressure field, cylinder Re 20 (wake zoom)

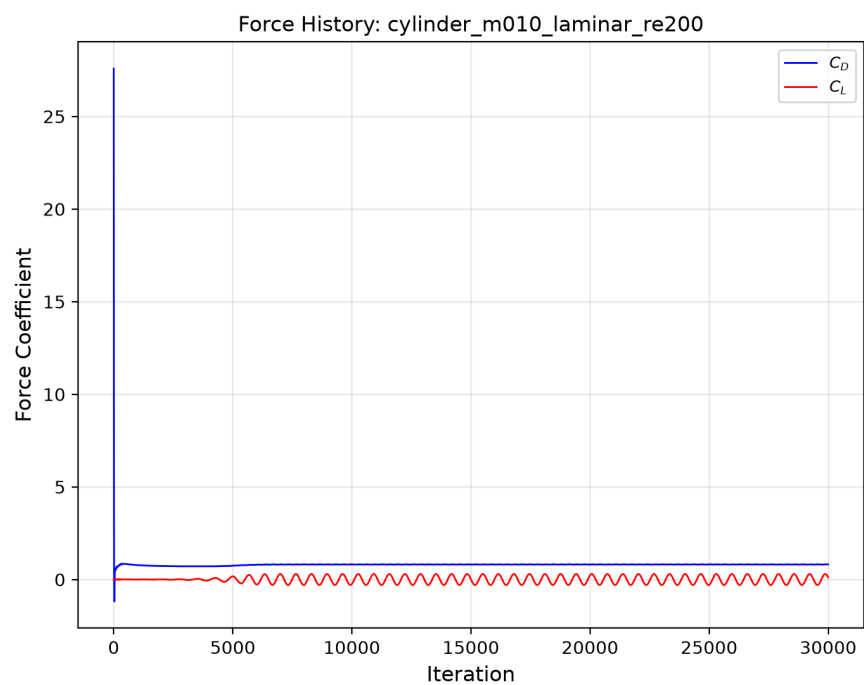


Figure 12: Force history, cylinder Re 200 (transient)

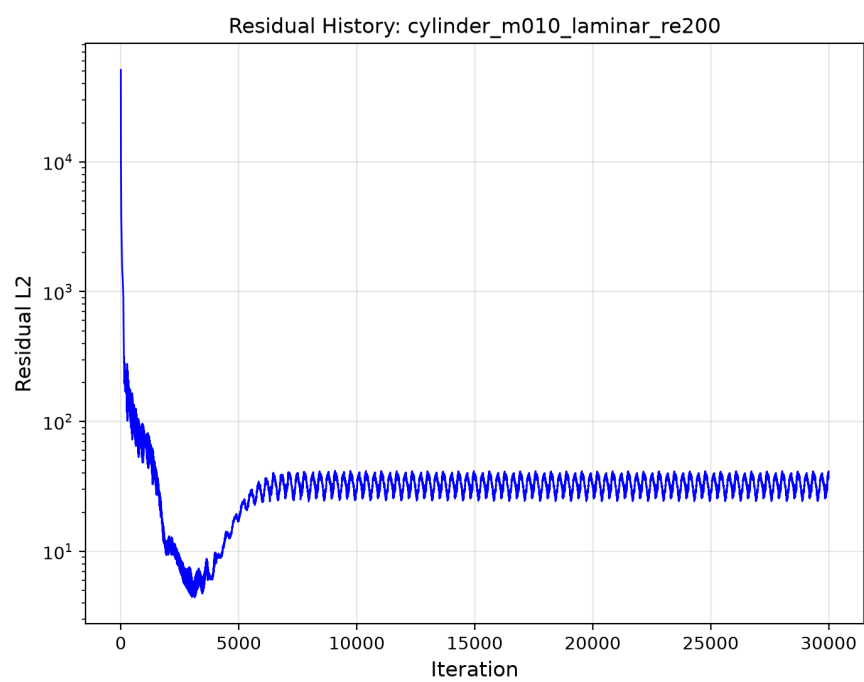


Figure 13: Residual history, cylinder Re 200

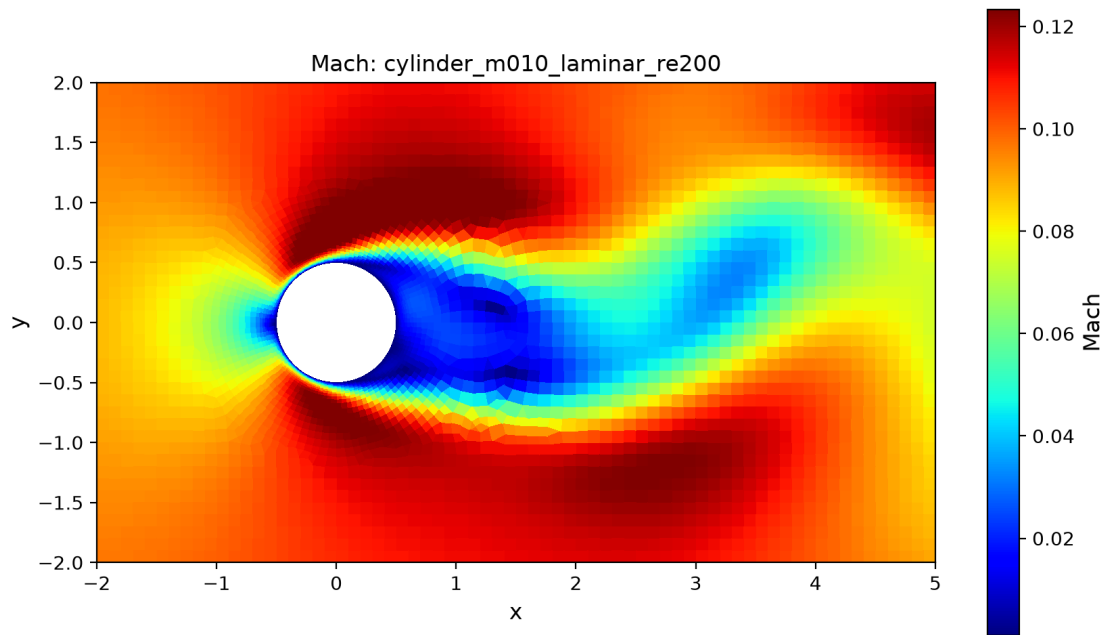


Figure 14: Mach field, cylinder Re 200 (wake zoom)

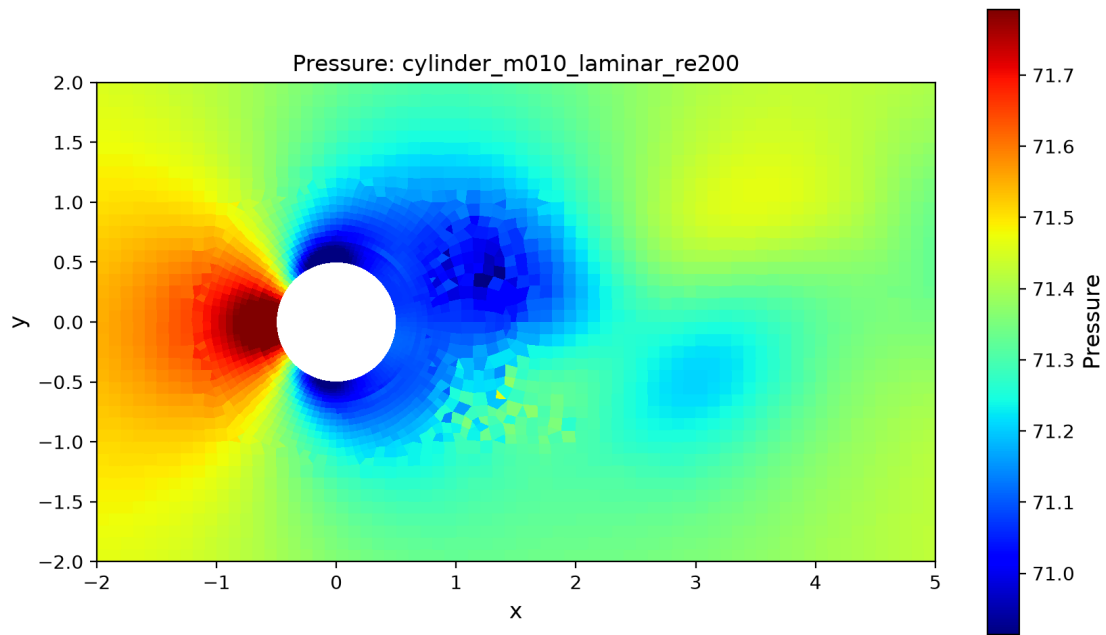


Figure 15: Pressure field, cylinder Re 200 (wake zoom)